

Mathematical Analysis I (Math I)

Fall 2026

Week 1 – Intro to course & Numbers

- Math, Math Analysis, Calculus
- How important are Math to engineering jobs?
- Why Math Analysis/Calculus is important to Engineering?
- Numeral (numeration) systems
- Positional representation of numbers
- Scientific and Engineering notations, SI prefixes
- Number systems: Natural, Integers, Rational, Real, Complex

Math, Math Analysis, Calculus

More on Mathematics, Mathematical Analysis and Calculus:

Mathematics is the broadest discipline, encompassing many fields such as algebra, geometry, probability, statistics, discrete mathematics, optimization, and mathematical analysis.

Mathematical Analysis forms one of its fundamental branches, providing the rigorous theory of continuous functions, limits, derivatives, integrals, and infinite processes. It provides the mathematical foundation for understanding how systems evolve over space and time. From the motion of planets to the training of artificial intelligence models, Mathematical Analysis offers the language and tools needed to describe, predict, and optimize continuous phenomena.

Calculus is the most application-oriented part of Mathematical Analysis. Differential calculus studies rates of change, slopes, optimization and approximation, while integral calculus studies accumulation, areas, volumes and total effects. Calculus is therefore closely related to mathematical analysis but normally emphasizes calculations, interpretation and applications more than formal proofs.

Note: In this course, we may use the terms “mathematical analysis” and “calculus” indistinguishably.

A few more things about Calculus...

Unlike mathematics as a whole, which studies arbitrary abstract structures, **calculus focuses on continuous structures**. Before calculus, mathematics was largely about static objects: numbers, triangles, equations, algebraic expressions.

Calculus introduced something fundamentally different: **Processes**. Instead of asking "What is this object?", calculus asks "**How is this object changing?**" and "**What accumulates as it changes?**" That's an enormous conceptual leap. As **nature changes continuously** (at least to a useful approximation), and real-world engineering systems involve continuous motion, continuous temperature, continuous voltage, continuous stress, etc., **calculus provides the math to analyze the continuous world**. [\[see, at the end of this file, a very recent result on the limitations of this statement\]](#)

Calculus addresses two fundamental questions...

1. How fast is something changing?

This leads to the *derivative*. Examples: velocity, acceleration, heat flow, electric current, learning rate, population growth. The derivative measures **local change**.

2. How much has accumulated?

This leads to the *integral*. Examples: distance traveled, energy consumed, mass, electric charge, total cost. The integral measures **global accumulation**.

... and unifies change and accumulation. We are going to discuss this later in Math I, when we will present the Fundamental Theorem of Calculus, which says that *the rate of accumulation equals the accumulated rate*.

Why Calculus is important

Calculus studies infinitesimal change and connects local-to-global reasoning

The key innovation was thinking locally. Instead of analyzing an entire curve, calculus asks "What happens at this instant?" The derivative is essentially an infinitely local measurement. The integral reconstructs the global picture from infinitely many local contributions.

Examples of how calculus connects local behavior with global behavior.

Local slope → Entire function

Local stress → Bridge stability

Local heat flow → Temperature distribution

Local learning rule → Neural network intelligence

The importance of MATH for engineering jobs

*Table: Ranking top knowledge domains in main engineering fields (based on data from O*NET)*

Materials & Nanotechnology	Electrical & Computer	Power & Energy	Industrial Eng. & Management
Engineering and Technology	Engineering and Technology	Engineering and Technology	Engineering and Technology
Mathematics	Computers and Electronics	Design	Design
Physics	Design	Mathematics	Mechanical
Design	Mathematics	Physics	Mathematics
Computers and Electronics	Physics	Computers and Electronics	Computers and Electronics
Chemistry	Mechanical	English Language	Production and Processing
English Language	English Language	Mechanical	Education and Training
Mechanical	Production and Processing	Administration and Management	Physics
Production and Processing	Education and Training	Chemistry	English Language
Education and Training	Administration and Management	Education and Training	Administration and Management

Numeral representation of numbers



10

2026

$$2026 = 2 \times 10^3 + 0 \times 10^2 + 2 \times 10^1 + 6 \times 10^0$$



$$2026 = 1 \cdot 2^{10} + 1 \cdot 2^9 + 1 \cdot 2^8 + 1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0.$$

Therefore, its binary representation is

$$2026_{10} = 11111101010_2.$$

Non-positional number representations (example)

Roman numerals ¶

Roman numerals use symbols such as

$$I = 1, \quad V = 5, \quad X = 10, \quad L = 50, \quad C = 100.$$

For example,

$$13 = XIII.$$

Roman numerals are useful for labels, dates and numbering sections, but arithmetic is considerably less convenient than in positional notation.

Roman numerals (cont'd)

2021 = *MMXXI*

and

9999 = *MMMMMMMMCMXCIX*

which is too lengthy.

However, it can become much more lengthy if we add one more decimal digit to the 9999 number:

99998 =

MMMMMMMMMMMMMMMMMMMM

MMMMMMMMMMMMMMMMMMMM

MMMMMMMMMMMMMMMMMMMM

MMMMMMMMMMMMMMMMMMMM

MMMMMMMMMMMMMMMMMMMM

MMMMMMMMCMXCVIII

Engineering notation and SI prefixes

SI prefixes are word parts added to base units in the International System of Units ([SI](#)) to represent powers of ten. They make very large or very small measurements easier to read and write.

Power	Prefix	Symbol	Example
10^{12}	tera	T	terawatt, TW
10^9	giga	G	gigahertz, GHz
10^6	mega	M	megawatt, MW
10^3	kilo	k	kilovolt, kV
10^0	—	—	volt, metre, second
10^{-3}	milli	m	millimetre, mm
10^{-6}	micro	μ	microampere, μA
10^{-9}	nano	n	nanometre, nm
10^{-12}	pico	p	picofarad, pF
10^{-15}	femto	f	femtosecond, fs
10^{-18}	atto	a	attosecond, as

Example

Speed of light ¶

The speed of light in vacuum is exactly

$$c = 299\,792\,458 \text{ m/s}.$$

Scientific notation:

$$c = 2.99792458 \times 10^8 \text{ m/s}.$$

Engineering notation:

$$c = 299.792458 \times 10^6 \text{ m/s}$$

or approximately

$$c \approx 300 \text{ Mm/s}.$$

Example

Power-engineering examples

A power station producing 500 000 000 W has an output of

$$5.00 \times 10^8 \text{ W}$$

in scientific notation and

$$500 \times 10^6 \text{ W} = 500 \text{ MW}$$

in engineering notation.

A transmission voltage of 220 000V can be written as

$$2.20 \times 10^5 \text{ V}$$

or, more naturally,

$$220 \times 10^3 \text{ V} = 220 \text{ kV}.$$

In-Class activity

The magnitude of the charge of an electron is

[illegible]

Express this magnitude in *Scientific Notation* and in *Engineering Notation*.

Solution of the exercise

Elementary electric charge

The magnitude of the charge of an electron is

$$e = 0.00000000000000000001602176634 \text{ C}.$$

Scientific notation:

$$e = 1.602176634 \times 10^{-19} \text{ C}.$$

Engineering notation:

$$e = 160.2176634 \times 10^{-21} \text{ C}.$$

In-Class activity (cont'd)

In-Class Activity:

Try to solve the following problems. Keep your solutions and check them with the answers provided later (and shown below). The in-class activity is to facilitate your learning; it will not be graded.

Exercise 1

Write 0.00000824 in scientific and engineering notation.

Exercise 2

Write $7.35 \times 10^8 \text{ Hz}$ in engineering notation.

Exercise 3

Write 0.0047 F using an SI prefix:

Solution

Exercise 1

Write 0.00000824 in scientific and engineering notation.

Scientific notation:

$$0.00000824 = \boxed{8.24 \times 10^{-6}}.$$

Since -6 is a multiple of 3, this is also valid engineering notation:

$$\boxed{8.24 \times 10^{-6} = 8.24 \mu}.$$

The prefix must be attached to an appropriate unit—for example,

$$8.24 \times 10^{-6} \text{ A} = 8.24 \mu\text{A}.$$

Solution

Exercise 2

Write $7.35 \times 10^8 \text{ Hz}$ in engineering notation.

Rewrite the exponent as a multiple of 3:

$$7.35 \times 10^8 = 735 \times 10^6.$$

Therefore,

$$\boxed{7.35 \times 10^8 \text{ Hz} = 735 \text{ MHz}}.$$

Exercise 3

Write 0.0047 F using an SI prefix:

$$0.0047 \text{ F} = 4.7 \times 10^{-3} \text{ F} = \boxed{4.7 \text{ mF}}.$$



significant figures

Significant figures are the digits in a number that express the precision and reliability of a measurement or calculation.  Britannica

You can learn more about the topic on [Significant figures | Britannica](#). 

Significant figures

9. Significant figures

Scientific notation clearly communicates the intended precision of a measurement.

For example,

$$3 \times 10^5$$

has one significant figure, while

$$3.000 \times 10^5$$

has four significant figures. Although both have the same nominal value, the second suggests greater measurement precision.

Suppose a length is measured as

$$L = 2.35 \text{ m.}$$

This value has three significant figures. Writing

$$L = 2.350000 \text{ m}$$

would incorrectly suggest a much more precise measurement unless that precision was actually available.

One more point to discuss...

10. Calculator and computer notation

Calculators and programming languages often use **E** or **e** notation:

2.99792458E8

means

$$2.99792458 \times 10^8.$$

Similarly,

1.602176634e-19

means

$$1.602176634 \times 10^{-19}.$$

Here, **e** means “multiply by 10 raised to the following exponent.” It does **not** mean Euler’s number $e \approx 2.71828$.